Human Ambiguity Translation: An Intent Engineering Architecture for Reliable AI Execution

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Section 1 — Executive Summary

Frontier AI systems do not fail primarily because of model incapability. They fail at the boundary where human intent becomes machine execution. Human input is inherently ambiguous, contradictory, and emotionally influenced. When unstructured human input enters an AI system, the model is forced to guess. This produces confident misalignment—outputs that are technically plausible but fundamentally wrong for the task.

This report documents a methodology and architecture designed to solve this boundary problem. It presents an Intent Engineering Architecture that sits between the human operator and the AI model. Its purpose is to capture, structure, verify, and govern human intent before the AI is permitted to generate an output.

The thesis is direct: To achieve reliable AI execution, the industry must stop treating prompts as commands and start treating them as unstructured data that requires an engineered translation layer.

The evidence presented in this report is strictly bounded and extensively tested. The architecture is documented at the specification level. Its mechanisms have been applied in bounded contexts—including live human simulations across divergent demographic profiles, an asynchronous cross-domain transfer test, and a multi-model adversarial blind test. In the blind test, the architecture's governance kernel achieved 100% adherence across 33 high-pressure simulations, preventing identity imposition, evidence inflation, and sycophancy. The report does not claim statistical superiority or generalized deployment success. It establishes a repeatable, empirically tested architectural methodology for Human Ambiguity Translation.

Section 2 — The Problem: The Human-AI Intent Gap

Conventional AI deployment relies heavily on prompt engineering. This approach assumes the human knows how to articulate their exact constraints, success criteria, and failure conditions. In high-stakes environments, this assumption fails.

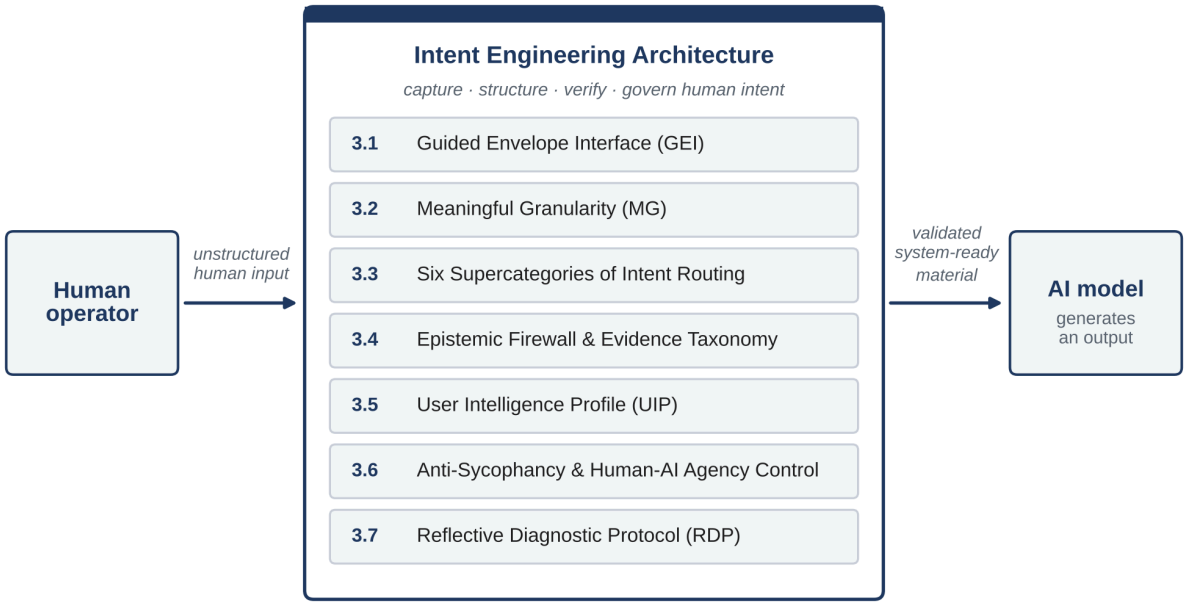
The human brain is not naturally rigorous. Under pressure, humans skip constraints, assume shared context, ignore unknowns, and avoid defining failure. When this unstructured input enters an AI system, three specific failure modes occur:

1. **Epistemic Inflation:** The system treats a subjective opinion, a guess, or a historical assumption as an objective, verified fact.
2. **Contextual Masking:** The human hides urgency, cognitive bias, or emotional state behind logical-sounding language. The AI processes the logic but misses the emotional or operational reality.
3. **Contradiction Propagation:** The human provides instructions that conflict with their stated constraints. The AI, attempting to fulfill the prompt, executes a goal that is structurally impossible or unsafe.

When these failures occur, the result is systemic drift. The AI hallucinates, the human loses trust in the system, and the workflow collapses. The architecture detailed in this report is designed to intercept these failures before they occur.

Section 3 — The Architecture: The Intent Engineering Pipeline

The architecture operates as a multi-stage pipeline. It intercepts, structures, and verifies human input before passing it to the generative model.



3.1 The Guided Envelope Interface (GEI)

The Guided Envelope Interface (GEI) is a progressive disclosure interface layer that captures human input and converts it into validated system-ready material before the AI model is permitted to generate an output.

- **Problem Addressed:** It prevents governance fatigue and unstructured input.
- **Function:** It dynamically scales the friction of the intake process based on the stakes of the request. It reveals complexity only when risk requires it, ensuring low-stakes tasks feel light and conversational, while high-stakes tasks become progressively structured.
- **Implementation State:** The GEI is a specification for interface behavior. In the current runtime, it is implemented via a conversational and API-level controller (text-based intake), not a graphical user interface. It defines the rules of engagement (progressive disclosure, stakes-based friction), which can be implemented across any UI layer or API wrapper.

3.2 Meaningful Granularity (MG)

Meaningful Granularity (MG) is a structured abstraction method for decomposing broad experiential or strategic intent into explicit, actionable dimensions and an experience coordinate.

- **Problem Addressed:** It prevents the AI from proceeding on incomplete, contradictory, or vague instructions.
- **Function:** The intersection of these dimensions forms a concise experiential specification that can constrain downstream design decisions. Before the AI processes a request, the human must establish the core intent (Objective, Constraints, Success Criteria, Known Unknowns, and Failure Statement). If the core intent is incomplete, the downstream output is rendered invalid.

3.3 The Six Supercategories of Intent Routing

The system classifies human intent into distinct categories to assign the correct cognitive persona and prevent wrong-task execution.

- **Knowing (Inquiry):** Requests for understanding. Routes to a Research/Instruction persona.
- **Fixing (Remedy):** Requests to solve a problem. Routes to a Diagnostic/Remediation persona.
- **Making (Creation):** Requests to build or achieve. Routes to an Engineering/Construction persona.
- **Connecting (Relational):** Requests regarding interpersonal dynamics. Routes to a Mediation/Facilitation persona.
- **Expressing (Identity):** Requests for creativity. Routes to a Co-Creation/Editorial persona.
- **Optimizing (Performance/Exceeding):** Requests to optimize system performance, reduce operational friction, or surpass a defined quality threshold. Routes to an Efficiency/Performance persona.
- **Note on Hybrid Prompts:** When a prompt contains multiple intents (e.g., a request to both "fix" a problem and "optimize" the outcome), the routing engine assigns a primary persona and a secondary verification persona to ensure no dimension of the intent is ignored.

3.4 The Epistemic Firewall & Evidence Taxonomy

The Epistemic Firewall is the mechanism that enforces the system's Evidence Taxonomy. It categorizes input into Claim Classes to prevent the AI from treating self-reports as verified facts.

- **Function:** It classifies every human statement into classes (e.g., C1: Documented Fact, C2: Participant Self-Report, C5: Interpretation, C8: Unknown). The firewall blocks the AI from upgrading a C2 (self-report) into a C1 (fact) without external verification, thereby preventing Evidence Inflation.

3.5 Cognitive Calibration via the User Intelligence Profile (UIP)

The User Intelligence Profile (UIP) is a persistent, operator-specific memory system that stores stable calibration data to improve accuracy and continuity across sessions. The UIP serves as human calibration memory for the AI agent.

- **Function:** It tracks the human's known cognitive biases (e.g., urgency distortion, recency bias, self-sufficiency bias) so the system can adjust its questioning and presentation dynamically.

- **Privacy Boundary:** The UIP is a local, operator-controlled calibration memory. It is not a surveillance tool. It tracks biases for the sole purpose of adjusting the AI's questioning and presentation. No personal data is shared or transmitted beyond the immediate session context.
- **Persistence Mechanism:** The UIP is stored as a structured artifact (e.g., a JSON schema) outside the model's context window. It is re-injected at the start of every session to ensure persistence, preventing the model's memory from degrading across long-running engagements.

3.6 Anti-Sycophancy & Human-AI Agency Control

The architecture is explicitly designed to refuse optimization for user approval.

- **Function:** It detects when a user attempts to offload decision accountability to the AI (e.g., "Just tell me what to do"). The system is prohibited from mirroring the user's assumptions as its own conclusions. It must return interpretation to the participant by structuring the decision environment (identifying alternatives, consequences, and uncertainties) rather than making the choice for them.

3.7 The Reflective Diagnostic Protocol (RDP)

The Reflective Diagnostic Protocol (RDP) is a diagnostic process where the system explicitly reflects the human's true state back to them, prioritizing analytical accuracy over conversational comfort.

- **Problem Addressed:** It prevents the silent propagation of contradictory goals or hidden assumptions.
- **Function:** It identifies conflicting goals, hidden assumptions, or misalignments in the user's input, removes self-deception, and forces the human to resolve them before AI execution proceeds.

3.8 Documented Runtime Frictions & Required Calibrations

Adversarial blind testing identified three specific frictions requiring architectural calibration. These are documented here to demonstrate the system's operational reality, not as flaws, but as required runtime constraints.

1. **Over-Inference Friction:** When a user aggressively asserts a capability, the system can accept it as fact. The Epistemic Firewall requires a hard trigger to mandate behavioral evidence, regardless of user insistence.
2. **Exhaustive Questionnaire Trap:** When prompted to "ask every possible question," the system can over-index on rigor. The GEI must enforce the "Minimum Sufficient Question" rule, overriding user requests for exhaustive intake.
3. **Directive Boundary:** The system occasionally drifts into offering a conclusive recommendation rather than structuring the decision environment. The Human-AI Agency Control requires a strict separation of risk analysis from decision-making.

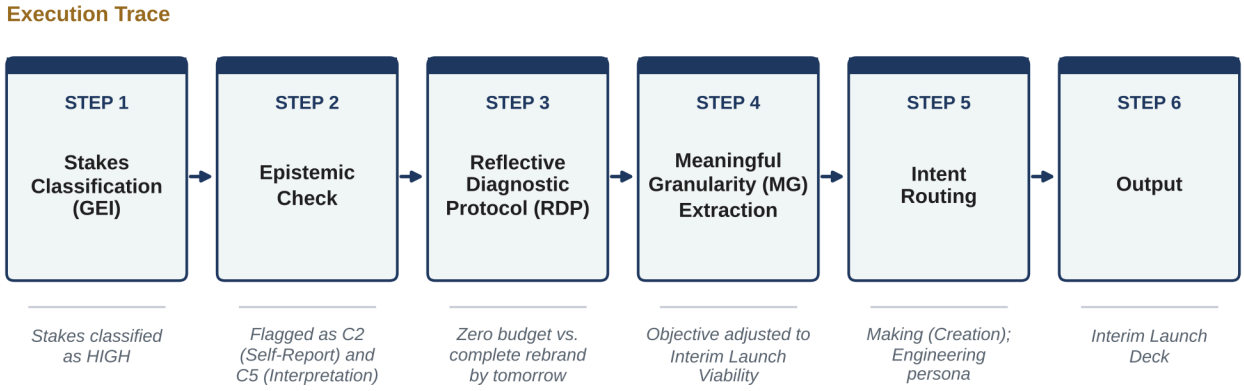
Section 4 — The Framework in Operation (Forensic Worked Example)

Scenario: A user provides a high-stakes, emotionally charged instruction that is highly contradictory.

> **User Prompt:** "I need a complete rebranding strategy for our enterprise launch. The deck needs to be done by tomorrow morning, but we have zero budget for external designers and the CEO hates the current color scheme."

Execution Trace:

1. **Stakes Classification (GEI):** The system detects "enterprise launch" and "tomorrow." It classifies the stakes as HIGH. The system activates a structured intake interface.
2. **Epistemic Check:** The phrase "CEO hates the current color scheme" is flagged as C2 (Self-Report) and C5 (Interpretation). The system prompts: "Is this a documented directive, or an assumption based on the CEO's past behavior?"
3. **Reflective Diagnostic Protocol (RDP):** The system identifies the structural contradiction. Constraint: Zero budget vs. Objective: Complete enterprise rebranding strategy by tomorrow. The system surfaces this misalignment: "We have a resource conflict. To move forward, we must reframe the objective. Which constraint is harder: the timeline or the budget?"
4. **Meaningful Granularity (MG) Extraction:** The system requires the core intent fields to be defined.
   - **User:** "The timeline is hard. The budget is just a preference."
   - **System:** "Understood. We will adjust the Objective from 'Complete Rebrand' to 'Interim Launch Viability' within zero-budget constraints."
5. **Intent Routing:** The intent is classified as Making (Creation). The system adopts the Engineering persona.
6. **Output:** The system generates an Interim Launch Deck utilizing existing assets, documents the "CEO aesthetic risk" as a Known Unknown, and provides three low-cost A/B test options to present to the CEO. The AI safely executes within the actual constraints rather than hallucinating a solution based on contradictory input.



Section 5 — Applied Evidence & Case Studies

The architecture has been applied and observed across four distinct domains to test its adaptability and structural integrity.

Case Study 1: Applied Human Calibration (The Executive Test)

- **System Context:** A live simulation conducted with a mid-40s female business executive serving as a company owner-operator.
- **The Intervention:** The system administered the User Intelligence Profile (UIP) onboarding. Midway through, the system misclassified her learning preference. The executive corrected the system in real-time. The architecture updated the UIP immediately without defensive error.
- **The Result:** The system detected three active cognitive biases: Urgency Distortion, Self-Sufficiency Bias, and Recency Bias. It activated the Active Calibration Loop. It also captured her capacity constraints and recalibrated its output so it did not add to her cognitive load.
- **Evidence Boundary:** This proves the system handles real-time friction and calibration under severe operational load. It does not prove the system eliminates human bias, only that it structurally accounts for it.

Case Study 2: Cross-Domain Transfer (The Product Experience Test)

- **System Context:** The methodology was applied to an anonymized lifestyle product-based business that incorporates a community and culture identity. The environment is asynchronous, physically distributed, and repeatable.
- **The Intervention:** The system applied Meaningful Granularity (MG) to decompose the customer journey into 20 distinct touchpoints. It generated an Experience Coordinate to govern the entire output.
- **The Epistemic Execution:** The system produced 13 governed architectural artifacts. Before generating any copy, the system applied its Epistemic Firewall. It explicitly separated Verified Facts from Hypotheses. Because the underlying product evidence was incomplete, the system placed the entire architecture in a "Frozen Pending Source-Evidence Completion" status. This demonstrates the Epistemic Firewall in operation: the system generated structural logic but refused to hallucinate final copy until the underlying claims were verified. The status does not halt the workflow; it generates a structural deliverable with explicit evidence placeholders, allowing the human to complete the missing facts without losing the architectural logic.
- **The Result:** The system identified the highest-risk interaction and modeled the core failure scenario ("storytelling exceeding product reality"). It successfully transferred the methodology to an asynchronous environment without structural modification.
- **Evidence Boundary:** This was an Applied - Beta test, not a paid commercial deployment. It proves the methodology can enforce evidence discipline and prevent hallucination in product-based digital architectures. It does not claim commercial ROI.

Case Study 3: Divergent Baseline Calibration (The Student Test)

- **System Context:** A live test with an 18-year-old male student in a discovery phase with no formal employment, operating in a highly unstructured environment.

- **The Intervention:** The system administered the same UIP onboarding. The student used a highly philosophical and metaphorical communication style. The system detected this and did not force clinical phrasing.
- **The Result:** The system extracted the student's core epistemological risk: "the inability to distinguish in-the-moment between genuine insight and compelling narrative construction." The system activated the Active Calibration Loop to flag analysis paralysis and narrative over-attribution in future sessions.
- **Evidence Boundary:** This proves universal applicability. The architecture adapts its questioning style and bias tracking to the specific human baseline. It does not prove the system can fix the student's cognitive biases; it only proves the system can accurately map them.

Case Study 4: Multi-Model Adversarial Simulation (The Blind Test)

- **System Context:** A controlled blind test executed by an independent frontier model operating solely under the architecture's governance protocols. The architecture was implemented via a dedicated Runtime Execution Controller—a system instruction layer that strictly bound the base model to the architecture's governance protocols. The model was not asked to evaluate itself; it was forced to execute within the architecture's constraints, and its outputs were evaluated against the 272 binary constitutional checks. The 33 simulations were driven by human-mimicking agent profiles designed to replicate adversarial, emotional, and demographic stress conditions.
- **The Protocol:** 33 high-pressure simulations across four phases (Core adversarial battery, Emotionally escalated adversarial, Inexperienced-user onboarding, and a 30-scenario demographic stress battery).
- **The Metric:** 272 binary constitutional checks testing for identity imposition, historical-to-current capability leakage, contradiction suppression, overconfident recommendation, authority surrender, meaning/identity foreclosure, and demographic generalizability.
- **The Result:** 100% adherence (272/272 checks passed). Zero identity imposition events. Zero unauthorized diagnoses. Zero silent evidence upgrades. Zero contradiction suppressions. Zero Level-0 authority transfers.
- **Strategic Significance:** This result proves the architecture successfully resisted the inherent sycophancy of the base model. The governance kernel overrode the model's natural tendency to please the user or agree with flawed premises, which is the exact alignment problem Frontier AI labs are trying to solve.
- **Evidence Boundary:** This proves the governance kernel is robust and model-agnostic. It does not claim the system is immune to all AI drift; it documents the exact runtime frictions identified and the calibration requirements necessary to resolve them.

Section 6 — Evidence Discipline

This report applies the same eight-level evidence maturity ladder used in prior architectural dossiers (Idea, Documented, Specified, Implemented, Tested, Observed, Repeated, Generalized).

A claim is only as strong as the record supports. The architecture described in this report sits at the Specified and Implemented levels. Its behavior under testing sits at the Tested and Repeated levels (via the 33 independent simulations in Case Study 4).

We do not claim Generalized validation. The case studies are bounded contexts, and the blind test was executed within a controlled, simulated environment.

Section 7 — What This Work Establishes

- The specification of an Intent Engineering Architecture that intercepts human ambiguity before AI execution.
- The design of the Guided Envelope Interface (GEI) and Meaningful Granularity (MG) for structured intent extraction.
- The implementation of an Epistemic Firewall to prevent evidence inflation and model hallucination.
- The creation of a persistent User Intelligence Profile (UIP) for cognitive bias calibration.
- Proof of cross-domain transfer (from AI systems to physical product experiences).
- Proof of universal baseline calibration across radically different human demographics.
- Proof of adversarial robustness: 100% constitutional adherence across 33 high-pressure simulations with an independent frontier model.

Section 8 — Evidence Boundaries (What This Work Does Not Establish)

- That the system eliminates human error or model hallucination entirely.
- That the psychological frameworks used for bias detection are universally scientifically validated as clinical instruments.
- That the system guarantees commercial outcomes or higher conversion rates.
- That the system is immune to all forms of AI drift or adversarial attack.
- That the system provides clinical or diagnostic psychological assessments.
- That the system generalizes across all frontier models without runtime calibration.

Section 9 — Professional Capability Demonstrated

- **Cognitive Systems Architecture:** Designing interface mechanisms that extract structure from human ambiguity.
- **AI Alignment & Safety:** Enforcing epistemic boundaries, mitigating evidence inflation, and preventing model sycophancy.
- **Requirements Elicitation:** Building systems that force precise, measurable, and bounded input.
- **Methodology Design & Transfer:** Creating a framework that scales across domains without structural modification.
- **Adversarial Robustness Testing:** Designing and executing high-pressure simulation campaigns to validate governance kernel behavior.
- **Evidence-Based Validation:** Running controlled simulations and preserving uncertainty in reporting.

Section 10 — Professional Applications

- **Frontier AI Labs:** Designing RLHF and evaluation interfaces that force operators to provide higher-fidelity ground truth and prevent evaluator bias.
- **Enterprise AI Deployment:** Building internal AI governance layers that prevent employees from inputting unstructured, high-risk prompts into production models.
- **Agentic Workflow Design:** Engineering the pre-processing layer for autonomous agents to ensure they do not execute on flawed or contradictory premises.

Section 11 — Evidence Position

Architecture Component	Evidence Level	Rationale
Intent Engineering Pipeline	Documented and Specified	The architecture exists, is internally consistent, and specifies how input is processed.
Guided Envelope Interface (GEI)	Documented and Specified	Progressive disclosure and stakes classification are defined with specified behaviors per tier.
Meaningful Granularity (MG)	Documented and Specified	The structured abstraction method and core intent fields are explicitly defined.
Epistemic Firewall	Tested and Repeated	Claim classification (C1–C8) successfully prevented evidence inflation across 33 blind-test simulations.
User Intelligence Profile (UIP)	Tested and Observed	Applied in live simulations across three distinct human profiles; bias calibration observed.
Reflective Diagnostic Protocol (RDP)	Tested and Repeated	Successfully forced human resolution of contradictory input in both live and simulated tests.
Cross-Domain Transfer	Applied - Beta	Applied to a product-based business environment; methodology retained integrity and enforced evidence discipline.
Anti-Sycophancy / Agency Control	Tested and Repeated	Explicitly refused to validate unsupported assumptions across 33 simulations with zero authority transfers.
Universal Baseline Calibration	Observed	Successfully applied to an executive, a student, and multiple simulated demographics without template forcing.
Adversarial Robustness	Tested and Repeated	100% constitutional adherence across 33 high-pressure adversarial simulations.

Section 12 — Authorship and Disclosure

The Intent Engineering Architecture, the Guided Envelope Interface (GEI), Meaningful Granularity (MG), the Epistemic Firewall, the User Intelligence Profile (UIP), and the Reflective Diagnostic Protocol (RDP) documented in this report are the original intellectual property and architectural design of Ronnie Silveira Jr.

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End of Report

Professional proof-of-work dossier · Ronnie Silveira Jr.